

HEUKSANDO, Korea, Republic of

WMO: 471690

Lat: 34.6872N

Lon: 125.4511E

Elev: 255

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.6	29.9	17.1	13.2	32.8	19.0	14.4	33.6	34.5	33.6	32.8	33.7	22.5	340	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.7	85.1	79.8	83.0	78.6	80.8	77.2	80.7	84.1	79.2	82.1	77.8	80.2	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	155.7	82.7	78.2	148.1	81.4	77.1	142.2	79.8	44.7	84.4	43.1	82.3	41.6	80.5	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.0	28.7	25.6	25.1	87.6	3.4	2.4	22.6	89.3	20.6	90.7	18.7	92.0	16.2	93.8		
(5)			23.1	82.0	3.0	2.7	20.9	83.9	19.2	85.5	17.6	87.0	15.4	88.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	38.1	38.9	43.9	51.4	59.9	67.1	73.6	77.0	70.8	62.8	52.7	42.5		
	DBStd	14.02	4.94	5.15	4.76	4.39	3.96	3.29	3.82	3.80	3.14	4.01	5.48	5.45		
	HDD50	1188	371	312	197	35	0	0	0	0	0	0	35	238		
	HDD65	3964	835	731	655	408	163	17	0	0	1	89	369	696		
	CDD50	3619	1	1	7	77	308	513	733	837	624	396	116	6		
	CDD65	921	0	0	0	0	6	80	268	372	175	20	0	0		
	CDH74	3850	0	0	0	1	14	61	1065	2432	274	3	0	0		
	CDH80	616	0	0	0	0	0	0	138	468	10	0	0	0		
(14) Wind	WSAvg	11.8	14.8	14.2	13.0	11.7	10.3	8.5	10.6	9.3	9.4	11.5	13.0	14.9		
(15) Precipitation	PrecAvg	44.6	0.9	1.3	2.1	3.5	3.8	6.0	8.5	8.5	4.5	2.0	2.0	1.3		
	PrecMax	65.1	2.0	3.3	4.9	8.5	8.3	14.4	13.8	24.5	10.3	5.7	5.4	3.4		
	PrecMin	28.2	0.2	0.2	0.3	0.8	0.7	1.9	3.3	1.6	0.6	0.1	0.1	0.1		
	PrecStd	8.7	0.5	0.9	1.2	2.0	1.9	3.3	3.0	6.0	2.7	1.5	1.4	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.8	55.8	59.4	67.6	75.6	77.9	86.6	88.9	81.5	74.2	66.9	58.8		
		MCWB	50.9	53.0	53.3	56.2	61.5	70.1	79.8	81.6	75.7	68.6	62.6	55.5		
	2%	DB	49.7	52.0	55.8	62.9	71.4	75.6	83.5	86.2	78.7	71.3	64.1	54.8		
		MCWB	46.1	48.7	51.1	55.3	61.1	69.3	78.9	80.4	73.6	66.6	61.0	51.2		
	5%	DB	46.7	48.6	53.5	60.3	68.6	74.0	81.2	84.5	76.9	69.2	62.1	52.3		
		MCWB	43.4	44.9	49.4	54.1	60.4	68.6	77.5	79.7	72.7	64.6	59.0	48.6		
	10%	DB	44.3	45.8	51.3	58.1	66.2	72.3	79.2	82.7	75.3	67.7	60.1	49.9		
		MCWB	40.4	42.4	47.3	53.0	59.2	68.0	76.3	78.5	71.3	63.4	56.6	45.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	54.3	55.3	60.3	66.6	73.7	82.0	84.2	77.4	71.5	63.7	55.6		
		MCDB	53.2	55.3	57.0	63.4	69.7	75.5	84.2	85.7	79.6	72.4	65.3	57.6		
	2%	WB	47.1	49.9	52.8	58.2	64.2	71.6	79.7	81.6	75.5	68.1	61.9	52.1		
		MCDB	49.2	51.3	54.7	60.7	67.6	73.6	82.5	84.7	77.4	69.8	63.7	54.3		
	5%	WB	43.9	45.7	50.6	56.2	62.7	70.1	78.0	80.2	73.9	66.2	59.8	49.3		
		MCDB	46.1	48.1	53.0	58.9	65.7	72.2	80.4	83.6	75.9	68.2	61.6	52.0		
	10%	WB	41.1	42.9	48.1	54.4	61.3	69.0	76.6	79.0	72.4	64.5	57.5	46.4		
		MCDB	44.2	45.4	50.5	57.0	64.4	71.1	78.7	81.9	74.3	66.9	59.6	49.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	6.7	8.7	9.6	9.8	8.5	7.3	7.7	6.5	6.1	6.4	6.0		
		MCDBR	9.3	12.1	13.2	12.6	13.1	10.5	9.6	10.0	8.5	8.1	8.7	9.5		
		MCWBR	8.6	10.6	10.8	9.2	8.0	6.4	5.8	5.8	6.0	6.3	7.9	9.2		
	5% WB	MCDBR	9.2	11.6	12.6	11.2	10.8	8.7	9.0	9.8	8.3	7.6	8.4	9.4		
		MCWBR	9.1	11.2	11.0	9.5	7.8	6.6	6.2	6.4	6.5	6.7	8.5	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.460	0.502	0.558	0.571	0.587	0.648	0.617	0.557	0.500	0.483	0.476	0.459		
	taud		1.940	1.869	1.758	1.783	1.802	1.752	1.876	2.041	2.120	2.023	1.972	1.945		
	Ebn at Noon		222	228	228	234	232	218	223	235	242	232	215	212		
	Edh at Noon		47	56	68	69	69	72	63	53	47	47	45	44		
(44) All-Sky Solar Radiation	RadAvg		672	948	1374	1576	1694	1527	1485	1582	1381	1178	797	590		
	RadStd		64	105	121	121	176	140	193	216	124	105	88	52		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220	+146
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+104	N/A

Nomenclature: See separate page